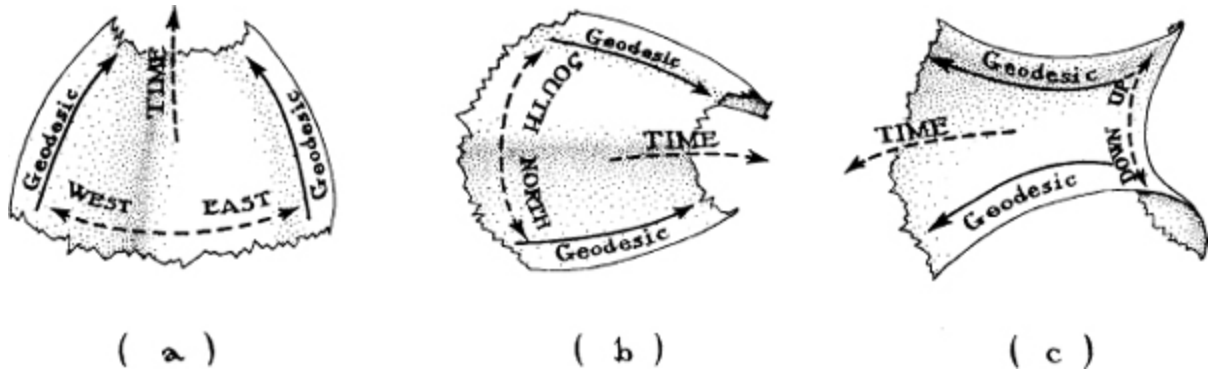
Einstein's law of spacetime warpage, the *Einstein field equation*, states that "mass and pressure warp spacetime." More specifically:

At any location in spacetime, choose an arbitrary reference frame. In that reference frame, explore the curvature of spacetime by studying how the curvature (that is, tidal gravity) pushes freely moving particles together or pulls them apart along each of the three directions of the chosen frame's space: the east–west direction, the north–south direction, and the up–down direction. The particles move along geodesics of spacetime (Figure 2.6), and the rate at which they are pushed together or pulled apart is proportional to the strength of the curvature along the direction between them. If they are pushed together as in diagrams (a) and (b), the curvature is said to be positive; if they are pulled apart as in (c), the curvature is negative.



(a) (b) (c)

Add together the strengths of the curvatures along all three directions, east–west [diagram (a)], north–south [diagram (b)], and up–down [diagram (c)]. Einstein's field equation states that *the sum of the strengths of these three curvatures is proportional to the density of mass in the particle's vicinity (multiplied by the speed of light squared to convert it into a density of energy; see Box 5.2), plus 3 times the pressure of matter in the particles' vicinity.*

Even though you and I may be at the same location in spacetime (say, flying over Paris, France, at noon on 14 July 1996), if we move relative to each other, your space will be different from mine and similarly the density of mass (for example, the mass of the air around us) that you measure will be different from the density that I measure, and the pressure of matter (for example, the air pressure) that we measure will differ. Similarly, it turns out, the sum of the three curvatures of spacetime that you measure will be different from the sum that I measure. However, you and I must each find